INCH-POUND

A-A-54930

31 January 1994

COMMERCIAL ITEM DESCRIPTION

ATTACHMENT, HEAD, HANDPIECE DENTAL, (Prophylaxis, Right Angle, Round Sleeve, Autoclavable)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers an autoclavable right angle, prophylaxis, dental handpiece head attachment to a handpiece assembly with a round nose, and for use with screw stem type polishing brushes and cups (See notes).

Salient characteristics:

Materials, hardnesses and finish. The attachment head components shall be fabricated of the materials and shall have the hardness values as specified.

Materials, hardness and finish.

COMPONENT (Part No.)	MATERIAL	ROCKWELL HARDNESS	FINISH
Rear End Housing (1)	303 Stainless Steel (UNS #S30300)	N/A	Wet Bead Blast
Pump Shaft Sub Assembly			
Pinion gear (2)	420 Stainless Steel (UNS #S42000)	Rc 48-51	Passivate
Pump shaft (3)	440C Stainless Steel (UNS #S44004)	N/A	Passivate
Bearing (4)	Phosphor Bronze B-2 (UNS #C54400)	N/A	N/A
Head Shell Assembly			
Bushing (5)	Phosphor Bronze B-2 (UNS #C54400)	N/A	N/A

AMSC N/A FSC 6520

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COMPONENT (Part No.)	MATERIAL	ROCKWELL HARDNESS	FINISH
Head Shell (6)	303 Stainless Steel (UNS #S30300)	N/A	Wet Bead Blast
Retainer Cap (7)	303 Stainless Steel (UNS #S30300)	N/A	Wet Bead Blast
Ring Gear (8)	420 Stainless Steel (UNS #S42000)	Rc 48-51	Passivate
Gear Barrel Shaft (9)	440C Stainless Steel (UNS #S44004)	Rc 54-58	Passivate
Lubrication Ring (10)	303 Stainless Steel (UNS #S30300)	N/A	Wet Bead Blast

Phosphor Bronze shall be in accordance with ASTM Designation B139 for composition only.

Stainless Steel shall be in accordance with ASTM Designation F899 for composition only.

Hardness shall be in accordance with ASTM E18 (Rockwell) or E92 (Vickers) and E140.

Design and dimensions. The attachment head shall be of the type known to the dental trade as a short sheath, round sleeve, right angle, prophylaxis handpiece. The attachment head shall be suitable for use with a round nose, air driven, handpiece assembly. The attachment head shall consist of a head assembly and back end assembly. The attachment head shall be provided with a screw stem socket threaded to receive (1-64 UNC) screw stem polishing brushes and (1-64 UNC) screw stem polishing cups.

The design and dimensions of the attachment head shall be in accordance with Figure 1. The prophylaxis attachment head shall be capable of being autoclaved and lubricated without disassembly. The attachment head shall have the retainer cap crimped into the head shell assembly to reduce penetration of prophylaxis pastes.

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Finish. The external surfaces of the attachment head shall be smooth. The finish shall be wet bead blasted except for the knurled portions.

Sterilization. The attachment head shall withstand a minimum of 62 steam autoclaves in a 14 day period, 5 cycles per day, without showing deterioration of components or breakdown of the attachment head. The attachment head shall be steam autoclaved at a temperature of 250-270°F (120-132°C) at 15-27 psi for 15-20 minutes.

After autoclaving the attachment head shall be:

1. Returned to room temperature.
2. Attached to a handpiece assembly.
3. Run at 3500 RPM for 15 minutes.
4. Checked at head and neck for heat rise (105°F maximum temperature allowed).

One cycle constitutes the washing, steam autoclaving and lubrication of the attachment head.

Runout. The runout of a true-running screw stem mandrel gage inserted in the attachment head shall not be more than 0.006 inch. A true-running test mandrel gage shall be screwed into the socket of the assembled attachment head and rotated slowly. A dial indicator sensitive to 0.0002 inch, equipped with a knife edge foot or other suitable equipment of equal accuracy shall be employed. The difference between the maximum and minimum readings for one full rotation of the gage, measured 1/8 inch from the screw socket, shall be reported as the runout.

Vibration and heat rise. The assembled and lubricated attachment head shall be tested by running the attachment head, containing a screw stem polishing brush, for 3 minutes at 1500 ± 100 revolutions per minute against an acrylic disc under a 4 ± 1/10 ounce load. The attachment head shall be tested for excessive vibration when hand held and excessive heat rise (above 105°F) with a pyrometer thermocouple fastened against the head housing and against the back end housing.

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Drive train smoothness and backlash. The assembled and lubricated attachment head shall be tested by spinning the back end drive shaft through at least two full revolutions, both clockwise and counterclockwise, rising finger pressure only as applied by the thumb and index finger. Any detectable binding, slipping, backlash, or irregular mechanical interference shall cause the attachment head to be rejected.

End play. The end play of the screw stem portion of a rubber cup inserted in the attachment head shall be no less than 0.005 inch. A screw stem of a rubber cup shall be screwed into the socket of an assembled attachment head which shall be clamped securely in a vise. A dial indicator, sensitive to 0.0002 inch and equipped with a small ball tip shall be positioned on a flat surface of the screw stem perpendicular to the axis of the bur tube. The cup shall be pushed towards and away from the head of the attachment head so that end play can be measured. The difference between the maximum and minimum readings shall be reported as end play.

Identification marking. Each attachment head shall be permanently and legibly marked with the name or registered trademark of the manufacturer. Attachment head of foreign manufacture shall be marked with the name of the country of origin.

Service data. The contractor shall enclose, with each unit, two copies of each of one or more booklets which shall contain complete instructions for operation, maintenance and sterilization methods.

Lubricating devices. Each unit shall be supplied with a lubricating nozzle and a tube of lubricating grease.

Workmanship. The attachment head shall be free from defects such as burrs, pits, feathers, sharp corners, jagged edges, and coarse grind marks which detract from its appearance or impair its serviceability.

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Unit. Each (EA). One attachment head, as specified, constitutes one unit. Each attachment head, together with a lubricating nozzle and a tube of lubricating grease, shall be packaged in a suitable unit carton.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one attachment head. Sampling shall be inspection level S-2 with an Acceptable Quality Level (AQL) of 1.0.

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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one attachment head. Sampling shall be inspection level S-2 with an AQL of 1.0.

Chemical analysis. In the absence of contractor certification for the raw material, a chemical analysis shall be conducted. The chemical analysis shall be performed by reagent solutions, combustion methods, nonemission methods, or by spectrochemical analysis. Any ASTM method shall be acceptable.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition,the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-01-160-8382	ATTACHMENT, HEAD, HANDPIECE DENTAL, Right Angle, Prophylaxis, Round Sleeve, Autoclavable, Screw Type

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NOTE1: The following National Stock Numbers and their respective Item Identifications are referenced by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-138-2308	HANDPIECE ASSEMBLY, DENTAL
6520-00-890-1565	BRUSH, POLISHING, HANDPIECE, DENTAL
6520-00-982-9377	CUP, POLISHING, DENTAL HANDPIECE

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity

DoD-MB

Agent:

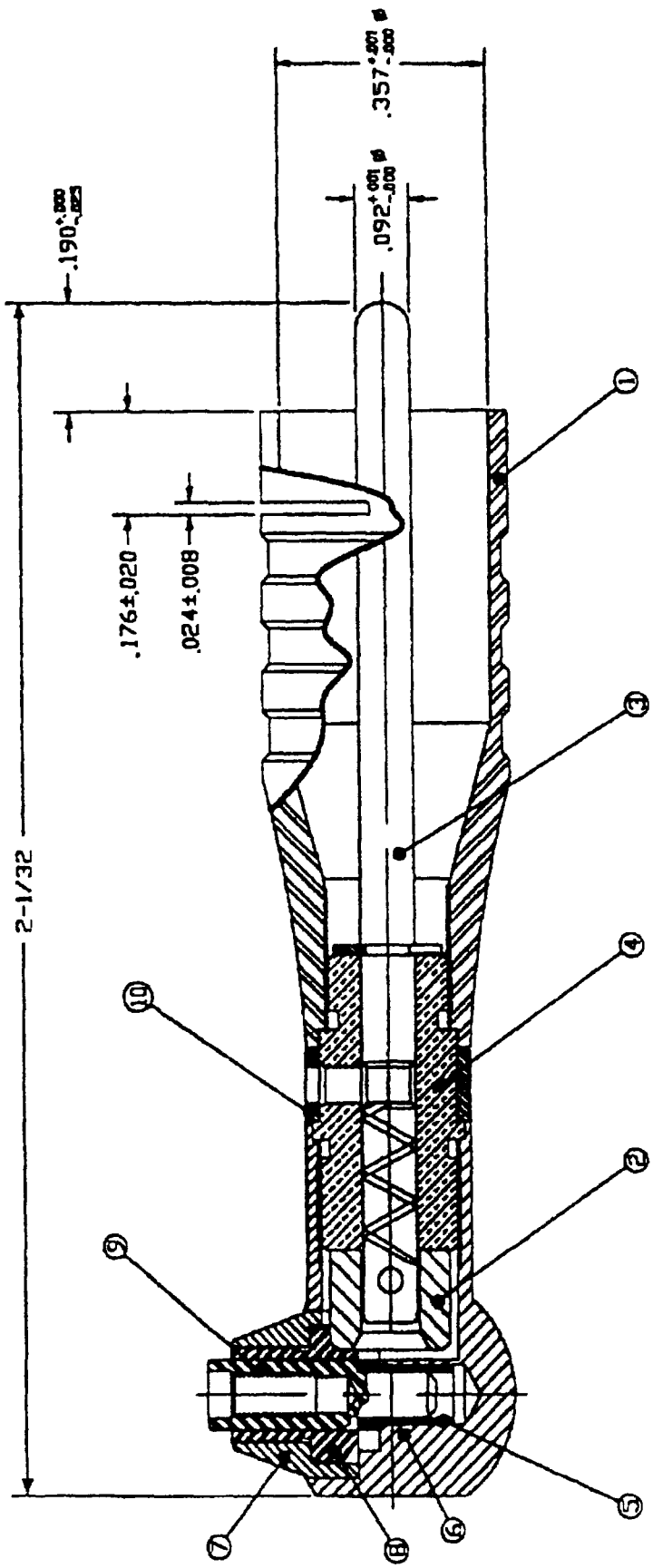
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6520-2555
Location: Intergraph/attachment

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NSN 6520-01-160-8382

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
TOLERANCES ON:
FRACTIONS ±1/32 DECIMALS ±.005 ANGLES ±2°

NO	COMPONENT
1	REAR END HOUSING
2	PINION GEAR
3	PUMP SHAFT
4	BEARING
5	BUSHING
6	HEAD SHELL
7	RETAINER CAP
8	RING GEAR
9	GEAR BARREL SHAFT
10	LUBRICATION RING

FIGURE 1 ATTACHMENT HEAD, HANDPIECE, DENTAL

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